Problem Statement

Project Title

Context-Aware Classification of Addiction-Related Commercial Advertisements using Deep Learning

---

1. Introduction

Commercial advertisements play a significant role in influencing public perception and consumer behavior. Products related to addiction, such as tobacco, alcohol, vaping products, hookah, betting platforms, and similar substances, are frequently promoted through commercial advertisements. At the same time, governments and health organizations release awareness campaigns to educate people about the harmful effects of addiction.

Existing AI-based systems can detect cigarettes, alcohol bottles, or tobacco products in images. Some systems can even identify brand logos or locate these objects using object detection techniques. However, these systems cannot understand the **intent** of an advertisement.

For example, two advertisements may contain an alcohol bottle. One may be a brand promotion encouraging alcohol consumption, while the other may be a government awareness campaign warning against alcohol abuse. Current AI systems generally treat both as the same because they only detect the object, not the message.

This project aims to develop a context-aware AI system that understands the purpose of addiction-related commercial advertisements.

---

2. Problem Statement

Current computer vision models answer only one question:

> **"Is an addictive product present in the advertisement?"**

They cannot determine:

- Is this advertisement promoting addiction?
- Is it simply showing the product without encouraging or discouraging it?
- Is it an awareness campaign?
- What message is being communicated to the audience?

Therefore, there is a need for an AI system capable of understanding the **context and intent** behind addiction-related commercial advertisements.

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3. Proposed Solution

We propose a deep learning framework that classifies addiction-related commercial advertisements based on their intent rather than simply detecting addictive products.

Instead of detecting only cigarettes or alcohol bottles, the model analyzes the complete advertisement and predicts one of the following categories.

Promotional

Advertisements that encourage, glamorize, or market addiction-related products.

Examples:

- Alcohol advertisements
- Tobacco advertisements
- Vaping product advertisements
- Hookah promotions
- Gutka
- Cigarates

Neutral

Advertisements where addiction-related products appear but the advertisement neither promotes nor discourages their use.

Examples:

- Retail product displays
- Product catalog advertisements
- Informational product showcases
- News reports featuring addictive products
- Brand announcements without persuasive messaging

Preventive

Advertisements intended to discourage addiction and promote healthy behavior.

Examples:

- Government awareness campaigns
- Anti-smoking advertisements
- Anti-alcohol campaigns
- NGO awareness advertisements
- Public Service Announcements (PSAs)
- Rehabilitation awareness campaigns

The objective is to understand **what message the advertisement is trying to communicate**, not merely identify the presence of addictive products.

---

4. Dataset Creation

Since no publicly available dataset exists for this specific problem, a custom dataset will be developed.

Step 1

Collect commercial advertisement videos from sources such as:

- YouTube
- Television commercials
- Government awareness campaigns
- NGO campaigns
- Official brand advertisements

Only commercial advertisements will be included in the dataset.

Step 2

Organize videos into three folders:

Videos/

Promotional/

Neutral/

Preventive/

Step 3

Automatically extract representative frames using OpenCV.

Instead of manually taking screenshots, Python scripts will extract representative frames from each advertisement.

Step 4

Review all extracted images manually.

Remove:

- Blurry images
- Duplicate images
- Low-quality frames
- Irrelevant frames

Step 5

Organize the final dataset.

Dataset/

Promotional/

Neutral/

Preventive/

The final dataset will consist of manually verified advertisement images.

---

5. Deep Learning Model

The prepared dataset will be used to train image classification models.

Baseline model:

- EfficientNet-B0

Additional comparison models:

- ResNet50
- ConvNeXt (optional)

The model will learn visual cues such as:

- Product branding
- Celebrity endorsements
- Health warnings
- Government logos
- Promotional design elements
- Educational messages
- Warning symbols

The final prediction will classify an advertisement into:

- Promotional
- Neutral
- Preventive

---

7. Project Workflow

Commercial Advertisement Videos

↓

Automatic Frame Extraction
(OpenCV)

↓

Image Cleaning

• Remove Blurry Images

• Remove Duplicate Images

↓

Manual Verification

↓

Final Dataset

↓

Deep Learning Model
(EfficientNet)

↓

Advertisement Classification

↓

Final Prediction

Promotional

Neutral

Preventive

---

8. Expected Outcomes

The project will produce:

1. Custom Dataset

A curated dataset of addiction-related commercial advertisement images categorized into Promotional, Neutral, and Preventive classes.

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2. Deep Learning Classification Model

A context-aware AI model capable of automatically understanding the intent of addiction-related commercial advertisements.

---

3. Research Publication

A research paper describing:

- Dataset creation methodology
- Model architecture
- Experimental evaluation
- Comparative analysis
- Public health applications

---

9. Applications

The proposed framework can be used by:

- Government agencies
- Public Health Organizations
- NGOs
- Advertising Regulatory Bodies
- OTT Platforms
- Social Media Platforms

Potential applications include:

- Automatic advertisement screening
- Public health monitoring
- Content moderation
- Regulatory compliance
- Advertisement analytics
- Policy enforcement

---

10. Novelty

Most existing research detects cigarettes, alcohol bottles, tobacco advertisements, or smoking scenes.

Our research moves beyond object detection by understanding the **intent** of addiction-related commercial advertisements.

Instead of asking:

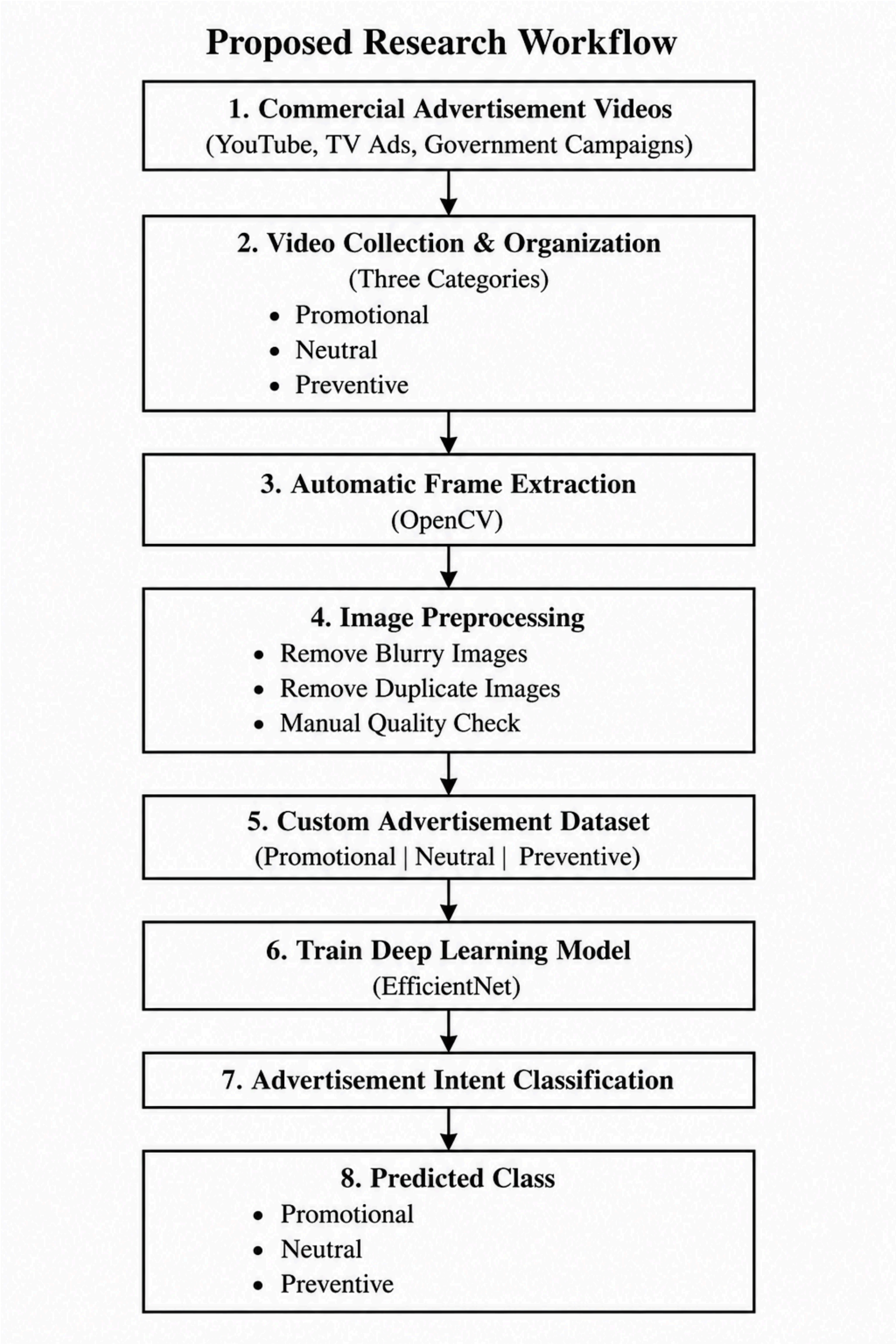
> **"Does this advertisement contain an addictive product?"**

our system asks:

> **"Is this advertisement promoting addiction, presenting information in a neutral manner, or spreading awareness?"**

This transition from object detection to **context-aware advertisement understanding** is the primary contribution and novelty of our work.

FLOW


Proposed Research Workflow
1. Commercial Advertisement Videos
(YouTube, TV Ads, Government Campaigns)
2. Video Collection & Organization
(Three Categories)
• Promotional
• Neutral
• Preventive
3. Automatic Frame Extraction
(OpenCV)
4. Image Preprocessing
• Remove Blurry Images
• Remove Duplicate Images
• Manual Quality Check
5. Custom Advertisement Dataset
(Promotional | Neutral | Preventive)
6. Train Deep Learning Model
(EfficientNet)
7. Advertisement Intent Classification
8. Predicted Class
• Promotional
• Neutral
• Preventive

Literature Review

Paper 1

Paper Title

The Promise of Zero-Shot Learning for Alcohol Image Detection: Comparison with a Task-Specific Deep Learning Algorithm

Published in **Scientific Reports (Nature), 2023**

---

1. Problem Statement

The authors observed that alcohol advertisements and alcohol-related content are increasing rapidly on social media, movies, online advertisements, and digital platforms. Research has shown that greater exposure to alcohol content increases alcohol consumption, especially among young people.

Currently, researchers manually analyze thousands of images to measure alcohol exposure, which is extremely time-consuming and expensive. Therefore, an automatic AI-based solution is required.

---

2. Objective of the Paper

The main objective is to compare two different approaches for detecting alcohol-related images:

Method 1

ABIDLA2

- A supervised deep learning model
- Specifically trained to detect alcohol images
- Requires a large labeled dataset

Method 2

CLIP (Zero-Shot Learning)

- A pretrained foundation model
- Does not require additional training on alcohol images
- Uses text prompts to classify images

The authors wanted to determine whether CLIP could achieve performance comparable to a specialized deep learning model.

---

3. Dataset Used

The paper used the ABD-2023 dataset.

Dataset details:

- Total training images: 164,671
- Validation images: 12,519
- Test images: 14,096

Eight categories were included:

- Beer Cup
- Beer Bottle
- Beer Can
- Wine
- Champagne
- Cocktails
- Whiskey/Cognac/Brandy
- Others

The authors also manually corrected mislabeled images before evaluation to improve dataset quality.

---

4. Methodology

The study compares two models.

A. ABIDLA2

- Supervised CNN
- Trained using labeled alcohol images
- Needs extensive annotation

B. CLIP

- Uses Zero-Shot Learning
- No retraining required
- Classifies images using text prompts

Example:

Instead of training on "beer bottle" images, CLIP is given prompts like:

- "Photo of a person drinking beer"
- "Photo of a bottle of beer on a table"

The image is compared with these prompts, and the closest match becomes the prediction.

---

5. Important Contribution

The authors introduced **Phrase Engineering**.

Instead of using a simple prompt like:

"Beer"

they used descriptive prompts such as:

"Photo of a person drinking a bottle of beer."

This significantly improved Zero-Shot Learning performance.

This shows that prompt design has a major impact on CLIP accuracy.

---

6. Experiments

Three classification tasks were evaluated.

Task 1

Identify the exact alcohol category.

Example:

Beer Bottle vs Beer Can vs Wine vs Champagne

Result:

ABIDLA2 performed better.

Task 2

Identify broader alcohol categories.

Example:

Beer

Wine

Spirits

Others

Result:

Both methods performed similarly.

Task 3

Binary Classification

Alcohol

vs

Non-Alcohol

Result:

CLIP slightly outperformed ABIDLA2.

This demonstrates that Zero-Shot Learning is highly effective for simple binary classification tasks.

---

7. Results

Main findings:

• ABIDLA2 achieved the best performance for fine-grained alcohol classification.

• CLIP achieved comparable performance for broader classification.

• CLIP slightly outperformed ABIDLA2 for binary alcohol detection.

• Descriptive prompts consistently performed better than simple class names.

---

8. Limitations

The paper only detects whether alcohol is present.

It cannot determine:

- whether the advertisement promotes alcohol,
- discourages alcohol,
- is an awareness campaign,
- or what the intention of the advertisement is.

The model focuses only on object detection and image classification.

---

9. Research Gap

Existing research answers:

"Does this image contain alcohol?"

Our research asks a different question:

"What message is this advertisement trying to convey?"

We classify advertisements into:

• Promotional

• Preventive

Thus, instead of only detecting alcohol, we analyze the advertisement's intent.

This is the main novelty of our work.

---

10. Learning

- Large labeled datasets improve supervised learning.
- Zero-Shot Learning is a good baseline.
- Prompt engineering improves CLIP performance.
- Image classification is feasible for addiction-related content.
- Existing work does not classify advertisement intent.

Therefore, our project extends existing research by introducing a new advertisement-intent dataset for addiction-related commercial advertisements.

this paper focuses on automatic alcohol image detection. The authors observed that manual analysis of alcohol-related images on social media and advertisements is very time-consuming, so they proposed using AI. They compared a task-specific deep learning model called ABIDLA2 with a foundation model called CLIP using Zero-Shot Learning. They evaluated both methods on a dataset containing more than 1.6 lakh labeled images across eight alcohol categories. They also introduced descriptive prompt engineering for CLIP and showed that it improves performance. Their results show that ABIDLA2 performs better for fine-grained classification, whereas CLIP performs similarly or even better for binary alcohol detection. However, the important limitation is that both methods only detect whether alcohol is present. They do not understand the intention behind the advertisement. This helped us identify our research gap. Instead of only detecting addictive products, our project will classify commercial advertisements into Promotional and Preventive categories based on the message they convey. That is the main difference between this paper and our proposed work.

Paper 2

Paper Title

Region Extraction Based Approach for Cigarette Usage Classification Using Deep Learning

Published by researchers from **IIT Roorkee (2021)**

---

1. Problem Statement

The authors start by explaining that smoking causes many health problems and is prohibited in many public places. However, it is difficult to monitor whether people are smoking everywhere using human supervision.

So, they wanted to develop an AI system that can automatically identify whether a person is smoking or not from an image.

---

2. Objective

The objective of this paper is simple:

Given an image of a person,

the AI should classify it into only two categories:

• Smoker

• Non-Smoker

If the person is a smoker, the system should also locate the cigarette in the image.

So this paper focuses on **smoking behavior detection**, not advertisement analysis.

---

3. Dataset Used

The authors used the **Mendeley Smoker Dataset**.

Dataset details:

- Total images = 2,400
- Around 1,200 smoker images
- Around 1,200 non-smoker images

The images contain different:

- backgrounds
- lighting conditions
- hand positions
- face angles
- environmental settings

They used an **80:20 train-test split** for evaluation.

---

4. Methodology

The paper proposes a three-stage pipeline.

Step 1 – Region Extraction

Instead of giving the whole image to the model, they first detect only the important regions.

These are:

- Face
- Hand

The reason is that cigarettes are very small objects and are usually present near the face or in the hand.

So the model ignores unnecessary background and focuses only on these important regions.

Step 2 – Classification

The extracted face and hand regions are passed to an ensemble of **ResNet-18** and **ResNet-34** models using transfer learning.

The model predicts:

Smoker

or

Non-Smoker

Step 3 – Cigarette Detection

Only if the image is classified as "Smoker", the system activates **YOLOv3** to detect the exact location of the cigarette.

If the image is classified as "Non-Smoker", YOLO is not executed.

This reduces unnecessary computation and also decreases false detections.

---

5. Key Idea

The biggest contribution of this paper is:

Instead of processing the complete image,

they first extract the important regions (face and hand),

then classify the image,

and only after that perform cigarette detection.

This makes the system faster and more accurate.

---

6. Results

The proposed method achieved:

- Accuracy = **96.74%**
- Precision = **95%**
- Recall = **98%**

The model also worked well in difficult situations such as:

- Dark backgrounds
- Side face
- Small cigarette visibility
- Multiple people in the image

The authors showed these challenging examples in their results section.

---

7. Limitations

Although the accuracy is high, the paper has several limitations.

The system only answers:

"Is this person smoking?"

or

"Where is the cigarette?"

It cannot answer questions like:

- Is this a tobacco advertisement?
- Is the scene promoting smoking?
- Is it an awareness campaign?
- What is the intention behind the image?

It only detects smoking behavior.

---

8. Research Gap

This paper focuses on **person-level smoking detection**.

Our project focuses on **advertisement-level understanding**.

Instead of checking whether someone is smoking,

we analyze the complete advertisement and classify it into:

- Promotional Advertisement
- Preventive Advertisement

Therefore, our work is different because we are trying to understand the message or intent of the advertisement, not just detect a cigarette.

---

9. Learning

- Small objects like cigarettes are difficult to detect.
- Region extraction improves model performance.
- Transfer learning is effective on small datasets.
- YOLO is useful for object localization.

- Existing research stops at smoking detection and does not analyze advertisement intent.

This supports our idea of building a new dataset and developing a context-aware advertisement classification system.

referred to this paper to understand how researchers handle small objects like cigarettes, how they design a detection pipeline using region extraction and YOLO, and what the current limitations are. It helped us identify that existing work focuses on smoking detection, whereas our research focuses on understanding the intent of addiction-related advertisements

Paper 3

Paper Title

Development of Framework for Detecting Smoking Scene in Video Clips

Published in **Indonesian Journal of Electrical Engineering and Computer Science (2019)**

---

1. Problem Statement

The authors observed that in India, whenever a smoking scene appears in a movie or TV show, a health warning has to be displayed. Currently, this process is done manually by editors, which is time-consuming and not practical for a large number of videos.

So, the authors wanted to develop an AI system that can automatically detect smoking scenes in videos and display warning messages without human intervention.

---

2. Objective

The main objective of this paper is:

- Detect smoking scenes in videos.
- Identify the presence of a cigarette.
- Automatically display a warning message like **"Smoking Kills"** or **"Smoking is injurious to health."**

The aim is automation of warning display, not understanding the meaning or intent of the scene.

---

3. Dataset Used

The authors did not use an existing public dataset.

Instead, they created their own dataset.

They:

- Collected videos containing smoking scenes.
- Extracted **5 frames per second** from those videos.
- Converted each frame into a 200 × 200 JPEG image.
- Manually annotated cigarette locations using LabelImg.

Final dataset:

- Around **660 training images**.
- Images contained different cigarette sizes, lighting conditions, backgrounds, and partially visible cigarettes.

---

4. Methodology

The pipeline is simple.

Step 1

Collect smoking videos.

↓

Step 2

Extract frames from the videos.

↓

Step 3

Annotate the cigarette using bounding boxes.

↓

Step 4

Train **Faster R-CNN with Inception ResNet v2**.

↓

Step 5

If a cigarette is detected, display a warning message automatically.

This is an object detection approach.

---

5. Why Faster R-CNN?

The authors compared different object detection models available in TensorFlow.

They selected **Faster R-CNN with Inception ResNet v2** because, although it is slower, it provides higher accuracy for detecting small objects like cigarettes.

---

6. Results

The model was tested on both images and videos.

For video testing:

- 10 videos were evaluated.
- The average detection accuracy was approximately **94%**.

The authors concluded that Faster R-CNN can effectively detect smoking scenes in videos.

---

7. Limitations

The system only answers one question:

"Is a cigarette present?"

If yes,

it simply displays a warning.

It cannot identify:

- Whether the scene is promoting smoking.
- Whether it is an anti-smoking campaign.
- Whether it is a commercial advertisement.
- What message the scene is trying to convey.

So, the understanding of context is completely missing.

---

8. Research Gap

This paper focuses on **automatic cigarette detection in videos**.

Our project is different.

We are not trying to detect cigarettes only.

We are trying to understand **what the advertisement is communicating**.

Our AI will classify commercial advertisements into:

- Promotional
- Preventive

based on the complete visual context.

---

9. Learning

From this paper we learned:

- Video frames can be converted into an image dataset.
- LabelImg can be used for annotation.
- Faster R-CNN performs well for cigarette detection.
- Creating a custom dataset from videos is a valid research approach.
- Existing work detects objects but does not analyze advertisement intent.

This paper also gave us confidence that creating our own dataset from advertisement videos is an accepted research methodology.

this paper proposes an AI framework to automatically detect smoking scenes in videos so that health warnings can be displayed without manual editing. The authors created their own dataset by collecting smoking videos, extracting five frames per second, and annotating cigarette locations using LabelImg. They trained a Faster R-CNN with Inception ResNet v2 model because it provides good accuracy for detecting small objects like cigarettes. The model achieved around 94% accuracy on video testing. However, the limitation is that it only detects whether a cigarette is present and then displays a warning. It cannot understand whether the scene is promoting smoking or discouraging it. This helped us identify our research gap. In our work, instead of only detecting cigarettes, we will analyze commercial advertisements and classify them as Promotional or Preventive based on the overall message they convey.

Paper 4

Paper Title

An Automated Framework for Detecting and Quantifying Alcohol-Related Content in Movies

Published in **International Journal of Computing and Digital Systems (2025)**

---

1. Problem Statement

The authors observed that alcohol is shown very frequently in movies, and research has proved that repeated exposure to such scenes can influence children and young viewers. Currently, regulatory bodies and researchers manually watch movies to identify alcohol scenes, but this process is very time-consuming, subjective, and not suitable for analyzing thousands of movies. Therefore, they wanted to build an AI system that can automatically analyze an entire movie and measure how much alcohol-related content it contains.

---

2. Objective

The objective of this paper is:

- Detect alcohol-related scenes in full-length movies.
- Measure how much alcohol appears in the movie.
- Express alcohol exposure as a percentage of the total movie duration.

Unlike simple object detection, they wanted to quantify alcohol exposure throughout the movie.

---

3. Dataset Used

There was no public dataset suitable for this task.

So the authors created their own dataset.

They:

- Collected Bollywood movie scenes.

- Extracted representative frames.
- Annotated alcohol bottles and glasses using Roboflow.

Final dataset:

- **758 images**
- **457 alcohol images**
- **301 non-alcohol images**
- Two object classes:
    - Alcohol Bottle
    - Alcohol Glass

The dataset was divided into:

- 73% Training
- 17% Validation
- 10% Testing

---

4. Methodology

The workflow is more advanced than previous papers.

Step 1

Input a full movie.

↓

Step 2

Split the movie into scenes using **PySceneDetect**.

↓

Step 3

Extract representative frames from every scene.

↓

Step 4

Detect alcohol bottles and glasses using **YOLOv8**.

↓

Step 5

Apply **Non-Maximum Suppression (NMS)** to remove duplicate detections.

↓

Step 6

Use contextual information like:

- Is it a bar or pub?
- Are children present?

to improve confidence.

↓

Step 7

Combine frame-level predictions into scene-level predictions.

↓

Step 8

Calculate:

Alcohol Content (%) = Alcohol Scene Duration / Total Movie Duration

This gives the final percentage of alcohol exposure in the movie.

---

5. Why did they use YOLOv8?

The authors selected YOLOv8 because:

- Fast inference
- Good accuracy
- Detects small objects like bottles and glasses
- Suitable for processing full-length movies

They also compared post-processing methods and found that **Non-Maximum Suppression (NMS)** was sufficient.

---

6. Results

The system performed well.

Main results:

- Final detection accuracy ≈ **90%**
- Fine-tuned YOLOv8 improved performance.
- The framework successfully analyzed **100 Bollywood movies**.
- Comedy movies showed the highest alcohol exposure compared to action, crime, and romance movies.

---

7. Limitations

Although this paper is very good, it still has limitations.

The system only detects:

- Alcohol bottle
- Alcohol glass

It measures **how much alcohol appears** in the movie.

But it cannot answer:

- Is the advertisement promoting alcohol?
- Is it an awareness campaign?
- Is the content encouraging addiction?
- Is it a surrogate advertisement?

It focuses only on alcohol exposure, not the intention behind the content.

---

8. Research Gap

This paper measures:

"How much alcohol is shown?"

Our research asks a different question:

"What message is the advertisement trying to convey?"

Instead of calculating alcohol exposure, we classify advertisements into:

- Promotional
- Preventive

Our goal is to understand the **intent of the advertisement**, not just detect alcohol-related objects.

---

9. Learning

From this paper we learned:

- How to build a custom dataset from videos.
- How to use PySceneDetect for video processing.
- Why YOLOv8 is suitable for object detection.
- How to combine frame-level predictions into scene-level analysis.
- How contextual information can improve accuracy.

Most importantly, we understood that existing work analyzes **movies**, while our research focuses specifically on **commercial advertisements** and their intent.

---

this paper proposes an automated framework to detect and measure alcohol-related content in full-length Bollywood movies. The motivation is that manually analyzing movies is very time-consuming, and repeated exposure to alcohol scenes can influence young viewers. Since no suitable dataset was available, the authors created their own dataset by collecting movie scenes and annotating alcohol bottles and glasses. Their pipeline first divides the movie into scenes using PySceneDetect, extracts representative frames, detects alcohol-related objects using YOLOv8, and then combines the results to calculate the percentage of alcohol exposure in the entire movie. They also use contextual information, like whether the scene is in a bar or whether children are present, to improve prediction accuracy. Their framework achieved around 90% accuracy and was tested on 100 Bollywood movies. However, the limitation is that it only measures alcohol exposure and cannot understand whether the content is promoting alcohol or discouraging it. This helped us identify our research gap. Our project will focus on commercial advertisements and classify them into Promotional or Preventive categories based on the message they convey, rather than only detecting alcohol-related objects."

---

this paper is the closest to our work because it also creates a custom dataset from videos and uses YOLOv8 for object detection. We learned how to design the data collection pipeline, process videos into frames, and perform scene-level analysis. However, their work is limited to measuring alcohol exposure in movies. Our work extends this idea by focusing on commercial advertisements and classifying their intent as Promotional or Preventive, which has not been addressed in this paper.

Paper 5

Paper Title

Automated Detection of Street-Level Tobacco Advertising Displays

Published as an **Authorea Preprint (2021)** by researchers from **NYU Center for Urban Science & Progress**.

---

1. Problem Statement

The authors observed that although tobacco advertising is restricted in television and newspapers, companies still heavily advertise tobacco products at retail shops and point-of-sale (POS) locations.

Currently, researchers manually visit stores or inspect thousands of street images to identify tobacco advertisements. This process is slow, expensive, and difficult to scale across an entire city.

Therefore, the authors proposed an AI-based system to automatically detect tobacco advertisements from Google Street View images and analyze how different communities are exposed to tobacco marketing.

---

2. Objective

The objectives of this paper are:

- Automatically detect tobacco advertisements in street-level images.
- Replace manual inspection with AI.
- Map tobacco advertisement locations across New York City.
- Measure youth exposure to tobacco advertising.
- Support public health policy and government regulations.

---

3. Dataset Used

The authors created their own dataset.

Data Collection Process:

- Used **Google Street View API**
- Collected images from licensed tobacco retailers across New York City.
- Captured **4 images** from every location (0°, 90°, 180°, 270°).
- Total collected images:
    - **42,776 street-view images**
- Images were manually annotated using **LabelImg** in **Pascal VOC XML** format.

To improve model robustness, they also applied image augmentation such as:

- Color balance
- Contrast adjustment
- Noise addition

This helped reduce overfitting and improved generalization.

---

4. Methodology

The workflow of the paper is as follows:

Step 1

Collect street-view images using Google Street View API.

↓

Step 2

Manually annotate tobacco advertisements using LabelImg.

↓

Step 3

Apply image preprocessing and augmentation.

↓

Step 4

Train **Faster R-CNN** object detection model.

↓

Step 5

Detect tobacco advertisements by generating bounding boxes.

↓

Step 6

Map detected advertisements geographically.

↓

Step 7

Combine advertisement locations with demographic information.

↓

Step 8

Generate heatmaps showing youth exposure to tobacco advertisements across New York City.

---

5. Why did they use Faster R-CNN?

The authors selected **Faster R-CNN** because:

- It is a state-of-the-art object detection algorithm.
- Detects objects along with their exact location.
- Generates accurate bounding boxes.
- Can distinguish tobacco advertisements from other street signs.

They experimented with different anchor sizes and found that the anchor scale **[4, 8, 16]** produced the best detection performance.

---

6. Results

The proposed system successfully detected tobacco advertisements in street-view images.

Main findings:

- Best Mean Average Precision (mAP): **0.62**
- Improved detection accuracy by **6%** compared to the baseline model.
- Successfully detected popular cigarette brands such as **Marlboro** and **Newport**.
- Generated geographic maps showing areas where children are most exposed to tobacco advertisements.

The system demonstrated that AI can effectively replace manual field surveys.

---

7. Limitations

Although the system performs well, it has several limitations.

It can:

- Detect tobacco advertisements.
- Locate advertisements geographically.

However, it **cannot** determine:

- Whether the advertisement promotes tobacco use.
- Whether it is an anti-smoking awareness campaign.
- The intention or message behind the advertisement.
- Whether the advertisement encourages addiction or discourages smoking.

It only detects the presence of tobacco advertisements.

---

8. Research Gap

This paper answers the question:

"Where are tobacco advertisements located?"

Our research answers a different question:

"What is the intention of the tobacco advertisement?"

Instead of only detecting tobacco advertisements, our project classifies commercial advertisements into:

- Promotional
- Preventive

Thus, our work moves from **object detection** to **intent understanding**, making it more useful for automated advertisement monitoring.

---

9. Learning

From this paper we learned:

- How to build a large custom dataset.
- How to collect street-view images using APIs.
- How to annotate images using LabelImg.
- How Faster R-CNN performs object detection.
- How to create heatmaps for advertisement analysis.

Most importantly, we understood that current research focuses only on detecting tobacco advertisements, whereas our project focuses on understanding the **message and intent** behind tobacco-related advertisements.

---

this paper proposes an automated system to detect tobacco advertisements from Google Street View images. The motivation is that manually identifying tobacco advertisements across an entire city is very expensive and time-consuming. Therefore, the authors collected more than 42,000 street-view images from licensed tobacco retailers in New York City using the Google Street View API. They manually annotated the tobacco advertisements using LabelImg and trained a Faster R-CNN object detection model. The system detects tobacco advertisements by placing bounding boxes around them and then combines these detections with demographic information to generate heatmaps showing youth exposure to tobacco advertising. Their best model achieved an mAP of 0.62 and successfully detected popular cigarette brands like Marlboro and Newport. However, the limitation is that the system only detects where tobacco advertisements are present. It cannot understand whether the advertisement is promoting tobacco or discouraging its use. This limitation helped us identify our research gap. Our project focuses on commercial advertisements and classifies them into Promotional or Preventive categories based on their intent rather than simply detecting tobacco-related objects.

> this paper is closely related to our research because it uses AI to detect tobacco advertisements from images and demonstrates how to build a custom dataset, annotate images, and train an object detection model. However, its focus is limited to detecting the presence and location of tobacco advertisements. Our research extends this idea by analyzing commercial advertisements and classifying them based on their intent—whether they promote tobacco use or spread preventive awareness. This shift from detection to intent classification is our main contribution."

Paper 6

1. Problem Statement

Many tobacco advertisements today are **hidden or indirect (covert advertising)**, especially on social media, videos, and online platforms.

Traditional AI models require **very large datasets** to detect smoking-related content, but there are very few publicly available smoking datasets.

The authors wanted to develop an AI system that can accurately detect smoking-related content even when only a **small amount of training data** is available.

---

2. Objective

The objectives of the paper are:

- Detect smoking content from both **images/videos and text**.
- Work effectively with **small datasets (few-shot learning)**.
- Detect hidden tobacco advertisements.
- Combine image understanding and language understanding into one AI system.
- Improve the model continuously using human feedback.

---

3. Dataset Used

Unlike many papers, they **did not create one huge dataset**.

Instead they combined multiple sources:

Image Dataset

- Public **Smoker vs Non-Smoker** dataset (Mendeley).
- Additional smoking advertisement videos downloaded from YouTube (Marlboro and Philip Morris ads).
- Video frames extracted every second.

Text Dataset

Since no Hungarian smoking dataset existed,

they generated one using:

- Smoking synonym dictionary
- ChatGPT-generated smoking-related paragraphs
- Manual annotation

Finally they created:

- 80 paragraphs
- Around **49,000 characters**
- Around **7,160 words** for text training.

---

4. Methodology

The complete pipeline is:

Input

Video OR Text

↓

Step 1

Identify input type.

↓

If Video

- Split video into frames.
- Resize images.

↓

Step 2

Use **CLIP-ViT-B-32 Multilingual** model.

This compares every image with the concept **"smoking"** using cosine similarity.

↓

Step 3

Discard images that clearly don't contain smoking.

↓

Step 4

Remaining images are classified using **EfficientNet-B5**.

↓

If Text

Text goes through

XLM-RoBERTa

to detect smoking-related words and entities.

↓

Step 5

Experts review incorrect predictions.

↓

Step 6

Corrected data is added back for retraining (Human Reinforcement).

---

5. Models Used

Instead of training from scratch, the authors used **pre-trained models**:

CLIP-ViT-B-32

- Multimodal model
- Understands both text and images
- Filters smoking-related frames

EfficientNet-B5

- Image classification
- Final smoking detection

XLM-RoBERTa

- Named Entity Recognition (NER)
- Detects smoking-related terms in text

Ensemble Learning

Final prediction combines CLIP and EfficientNet for better performance.

---

6. Results

Image Detection

- Final accuracy: **74%**

Text Detection

- Accuracy: **98%**
- F1-score: **91%** on the test dataset

The ensemble model performed better than using CLIP or EfficientNet alone.

---

7. Novel Contribution

The major contribution of this paper is:

- Works with **very small datasets**.
- Uses **multimodal AI** (image + text).
- Uses **ChatGPT** to generate training text.
- Uses **human reinforcement** to continuously improve performance.

This makes the system practical even when labeled data is limited.

---

8. Limitations

Although the system detects smoking well, it still has some limitations.

It:

- Detects whether smoking exists.
- Detects smoking-related words.
- Detects smoking scenes.

But it **cannot determine**:

- Whether the content promotes smoking.
- Whether it is an anti-smoking campaign.
- The intention behind the advertisement.
- Whether it glamorizes tobacco use or discourages it.

The authors themselves mention that adding **object detection** and better localization would improve the system further.

---

9. Research Gap

This paper answers:

> **"Does this image or text contain smoking?"**

Our research answers:

> **"What is the intent of this tobacco advertisement?"**

Instead of only detecting smoking content, our project classifies commercial tobacco advertisements into:

- **Promotional**
- **Preventive (Awareness)**

Thus, our work goes beyond detection and focuses on **intent understanding**, making it more useful for automated advertisement monitoring.

---

10. Learning

From this paper we learned:

- How to combine **image and text** in one AI pipeline.
- How to use **CLIP** for multimodal filtering.
- How **EfficientNet-B5** improves image classification.
- How **XLM-RoBERTa** can analyze smoking-related text.
- How to use **human feedback** to improve AI models over time.
- How to build a good system even with **limited training data**.

---

this paper proposes a multimodal deep learning system for detecting smoking-related content from both images/videos and text, especially when only a small amount of training data is available. The authors first identify whether the input is a video or text. For videos, they split them into frames and use the CLIP-ViT-B-32 model to filter frames that are semantically similar to smoking. These filtered images are then classified using EfficientNet-B5. For textual content, they use the XLM-RoBERTa model to identify smoking-related entities. They also introduce a human reinforcement mechanism where experts review incorrect predictions and feed corrected samples back into the training process. The system achieved 74% accuracy for image detection and 98% accuracy for text detection. However, the limitation is that it only detects smoking-related content and cannot determine whether the content promotes smoking or discourages it. This limitation directly motivates our research, where we focus on understanding the intent of commercial tobacco advertisements by classifying them as Promotional or Preventive rather than only detecting smoking content.

---

this paper is highly relevant because it demonstrates how multimodal AI can analyze both visual and textual tobacco-related content, even with limited data. We learned techniques such as CLIP-based multimodal filtering, EfficientNet image classification, XLM-RoBERTa for text analysis, and human-in-the-loop learning. However, the paper only identifies the presence of smoking content. Our work extends this by understanding the intent behind commercial tobacco advertisements, classifying them as Promotional or Preventive, which makes our system more suitable for advertisement regulation and public health monitoring.

Paper 7

Title

Image Processing for Public Health Surveillance of Tobacco Point-of-Sale Advertising: Machine Learning-Based Methodology (2021)

---

1. Problem Statement

Tobacco companies spend a large amount of money on **Point-of-Sale (POS) advertising**, such as posters, banners, and product displays inside retail stores.

Currently, monitoring these advertisements requires **manual store audits**, which are:

- Time-consuming
- Expensive
- Require trained personnel
- Difficult to perform regularly

The authors wanted to automate this process using **Computer Vision and Deep Learning**.

---

2. Objective

The main objectives are:

- Detect tobacco advertisements from retail store images.
- Identify the tobacco brand present.
- Detect the exact location of advertisements.
- Reduce manual inspection using AI.
- Support public health surveillance and policy enforcement.

---

3. Dataset Used

The researchers collected **real-world retail store photographs**.

Data Collection

- **410 tobacco retailers** in West Virginia

- **82 retailers** in Washington, DC
- Camera glasses were used to capture store images.

Total images collected:

- **86,683 images** (West Virginia)
- **13,264 images** (Washington, DC)

After cleaning:

- **694 clear images** for brand classification
- **843 Marlboro images** for object detection.

---

4. Data Preprocessing

The collected images were:

- Blurry images removed
- Duplicate images removed
- Images without advertisements discarded
- Images manually labeled by brand

Dataset split:

- **70% Training**
- **30% Testing**

Only **0.8%** of the collected images were useful because many photos did not contain visible tobacco advertisements.

---

5. Methodology

Step 1

Capture images inside retail stores.

↓

Step 2

Preprocess images

- Resize

- Clean
- Label

↓

Step 3

Brand Classification

Use **Inception V3**

Output:

- Detects which tobacco brand is present.

↓

Step 4

Object Detection

Use **YOLOv3**

Output:

- Detects exact advertisement location.
- Draws bounding boxes.

↓

Step 5

Evaluate model performance using:

- Accuracy
- mAP
- IoU.

---

6. Models Used

Inception V3

Purpose:

Image Classification

Detects:

- Marlboro
- Camel
- Newport
- Pall Mall
- Copenhagen
- Blu
- American Spirit
- Winston

It predicts the probability that each brand appears in the image.

YOLOv3

Purpose:

Object Detection

Detects:

- Exact advertisement location
- Bounding box coordinates
- Advertisement size
- Confidence score

YOLO identifies **where** the advertisement appears in the store image.

---

7. Results

Brand Classification

For **8 tobacco brands**, classification accuracy was **greater than 75%**.

Some examples:

- Copenhagen – **90.4%**
- Winston – **85.1%**
- Pyramid – **82.7%**
- Blu – **81.7%**
- Marlboro – **78.8%**
- Camel – **75.9%**

Object Detection

YOLOv3 achieved:

- **mAP = 0.72**
- **IoU = 0.62**

The model successfully detected multiple Marlboro advertisements and drew accurate bounding boxes around them.

---

8. Novel Contribution

This paper introduces a machine learning pipeline that can:

- Automatically detect tobacco advertisements in store images.
- Identify different tobacco brands.
- Localize advertisements using bounding boxes.
- Reduce the need for manual public health surveillance.

This demonstrates that AI can assist public health agencies in monitoring tobacco marketing.

---

9. Limitations

The paper has several limitations:

- Small labeled dataset.
- Object detection was performed mainly for **Marlboro** due to limited data.
- Detection accuracy decreases with different advertisement colors, sizes, lighting, and viewing angles.
- Detects **only the presence and location** of advertisements.
- Does **not analyze advertisement intent or promotional messaging**.
- Cannot classify advertisements as promotional or awareness campaigns.

---

10. Research Gap

This paper focuses on:

> **"Where is the tobacco advertisement?"**

and

"Which tobacco brand is present?"

Our project goes one step further by asking:

"What is the purpose or intent of the advertisement?"

Instead of only detecting tobacco logos, our system classifies commercial tobacco advertisements into:

- **Promotional**
- **Preventive (Awareness)**

Thus, our research extends beyond object detection to **context-aware intent understanding**, making it more suitable for automated advertisement moderation and public health monitoring.

---

11. Learning

From this paper, we learned:

- How to collect and preprocess real-world tobacco advertisement images.
- How **Inception V3** can classify tobacco brands.
- How **YOLOv3** performs advertisement localization with bounding boxes.
- How **mAP** and **IoU** evaluate object detection performance.
- The importance of transfer learning when working with limited datasets.
- The need to move beyond logo detection toward understanding advertisement intent.

---

this paper proposes a machine learning framework for automating the surveillance of tobacco point-of-sale advertisements inside retail stores. The researchers collected over 99,000 store images from West Virginia and Washington, DC, and after preprocessing selected high-quality images for training. They used the Inception V3 model for tobacco brand classification and YOLOv3 for object detection to identify the exact location of advertisements using bounding boxes. The model achieved more than 75% classification accuracy for eight tobacco brands, while YOLOv3 obtained an mAP of 0.72 and an IoU of 0.62 for advertisement localization. The major contribution is reducing manual store audits through AI-based surveillance. However, the system only detects tobacco brands and advertisement locations. It cannot understand whether an advertisement promotes tobacco use or discourages it. Our research addresses this limitation by focusing on intent-aware classification of commercial tobacco advertisements into Promotional and Preventive categories, making it more useful for automated content regulation and public health applications.

---

this paper provides the foundation for our work because it shows how deep learning models such as Inception V3 and YOLOv3 can detect tobacco brands and localize advertisements in images. We plan to build upon this idea by moving beyond logo detection to understand the intent of commercial tobacco advertisements. Instead of only identifying where an advertisement is located, our model will classify whether it is Promotional or Preventive, which is a higher-level semantic understanding required for public health monitoring.

Dataset builder Code

```
import cv2
import os

# ==========================================
# CONFIGURATION
# ==========================================

BASE_VIDEO_FOLDER = r"E:\Research Project\Videos"
BASE_OUTPUT_FOLDER = r"E:\Research Project\ExtractedFrames"

FRAME_INTERVAL_SECONDS = 1

# ==========================================

def extract_frames(video_path, output_folder, prefix, video_id):

    os.makedirs(output_folder, exist_ok=True)

    cap = cv2.VideoCapture(video_path)

    fps = cap.get(cv2.CAP_PROP_FPS)

    frame_interval = int(fps * FRAME_INTERVAL_SECONDS)

    frame_count = 0
    saved_count = 1

    while True:

        success, frame = cap.read()

        if not success:
            break

        if frame_count % frame_interval == 0:

            filename = f"{prefix}_{video_id}_F{saved_count:03}.jpg"

            save_path = os.path.join(output_folder, filename)

            cv2.imwrite(save_path, frame)

            saved_count += 1
```

```
        frame_count += 1

    cap.release()

    print(f"{video_id} --> {saved_count-1} frames extracted")


# ==========================================

for category in ["Promotional", "Preventive"]:

    video_folder = os.path.join(BASE_VIDEO_FOLDER, category)

    output_folder = os.path.join(BASE_OUTPUT_FOLDER, category)

    videos = os.listdir(video_folder)

    for video in videos:

        if not video.endswith(".mp4"):
            continue

        video_path = os.path.join(video_folder, video)

        filename = os.path.splitext(video)[0]

        parts = filename.split("_")

        prefix = parts[0]

        video_id = parts[1]

        extract_frames(
            video_path,
            output_folder,
            prefix,
            video_id
        )

print("\n===================================")
print("Dataset Generation Completed")
print("===================================")
```

Frame extractor

```
import cv2
import os

# -----------------------------
# Configuration
# -----------------------------
video_path = r"E:\Research Project\Videos\1968 Marlboro Cigarette
Commercial [0_4b7DTorXk].mp4"
output_folder = r"E:\Research Project\ExtractedFrames\Marlboro"

video_id = "0_4b7DTorXk"
prefix = "MARLBORO"

frame_interval_seconds = 2

# -----------------------------
# Create output folder
# -----------------------------
os.makedirs(output_folder, exist_ok=True)

# Open video
cap = cv2.VideoCapture(video_path)

fps = cap.get(cv2.CAP_PROP_FPS)
frame_interval = int(fps * frame_interval_seconds)

frame_count = 0
saved_count = 1

while True:
    success, frame = cap.read()

    if not success:
        break

    if frame_count % frame_interval == 0:

        filename = f"{prefix}_{video_id}_F{saved_count:03}.jpg"

        save_path = os.path.join(output_folder, filename)

        cv2.imwrite(save_path, frame)
```

```
        print(f"Saved {filename}")

        saved_count += 1

    frame_count += 1

cap.release()

print("\nDone!")
print(f"Total Frames Saved: {saved_count-1}")
```

Drive Link

https://drive.google.com/drive/folders/1_be8Vi0Won4zN2jbnfNctUWdRMNVpOud?usp=drive_link